Alignment ladder: exact local certificates and the remaining shared update

8 October 2026 — methodology review; no traffic simulation or model adoption

Conclusion. The unchanged uniform field fails a local compatibility test. Restriction-aware and continuous-wave descriptions have exact local constructions, including own motion. Neither a completed numerical update nor an applicable full continuum result has yet been supplied for all approved lane-transfer, node, record and reset interactions. This is not a theorem that the model cannot be constructed. A full arbitrary-data continuum theorem is not required if an admissible, coherent finite numerical model is qualified.

1. Quantities and the unchanged projection obstruction

Traffic density ρ and jam density κ have units equivalents/m. Free speed u, backward-wave magnitude w and restriction speed s have units m/s. Road flux f has units equivalents/s. The triangular diagram is

$$f(\rho) = \min\{u\rho, w(\kappa - \rho)\}, \qquad \rho_c = \frac{w\kappa}{u+w}, \qquad Q = u\rho_c. \tag{1}$$

Let $C(t)$ be a local cumulative incoming count (with a fixed gauge offset), and $N(t,x)$ the count remaining after subtracting traffic upstream of x:

$$N(t,x) = C(t) - \int_0^x \rho(t,\xi)\,d\xi, \qquad N_x = -\rho, \quad N_t = f(\rho). \tag{2}$$

The last identity is for a source-free segment with its correctly accounted upstream port. A restriction position $z_j(t)$ has count $h_j = N(t, z_j(t))$. Its speed $v_j = \dot{z}_j$ gives

$$\dot{h}_j = f(\rho) - v_j\rho = 0 \tag{3}$$

on the approved zero-relative-passing path. Records are effect metadata; they add no population to these integrals.

Recheck the earlier 30 m cell, $\Delta t = 1$, $u = 30$, $w = 5$, $\kappa = 3/20$, initial density $1/100$, free exit, sufficient input, and coincident old records with $h_1 = h_2 = 0$, desired speeds 5 and 15. The own fan at one second has $\rho = 3/40$ on $[0,5]$ and vacuum on $(5,30)$; positions are 5 and 15. The initial $3/10$ mass exits and $3/8$ enters. Consequently

$$m(1) = C(1) = \frac{3}{8}, \qquad N(1,5) = N(1,15) = N(1,30) = 0. \tag{4}$$

Uniform projection instead uses $\bar{\rho} = (3/8)/30 = 1/80$ and yields

$$\bar{N}(1,5) = \frac{5}{16}, \qquad \bar{N}(1,15) = \frac{3}{16}, \qquad \int_5^{15} \bar{\rho}\,dx = \frac{1}{8}. \tag{5}$$

Thus positions, old labels and uniform reconstruction cannot all hold. The parent CFL equality $u\Delta t = 30$ is satisfied. The example is local, not a reached metering witness or a proof against every CTM realization.

2. A new exact two-restriction own-field evolution

Use the same diagram in $[0, 30]$. At time zero the field consists of two congested blocks separated by vacuum. Let

$$r_1 = \frac{w\kappa}{w+5} = \frac{3}{40}, \qquad r_2 = \frac{w\kappa}{w+10} = \frac{1}{20}. \tag{6}$$

For $0 \le t \le 1/2$, prescribe only the external upstream demand needed to maintain the own admissible inflow $q_0 = 5r_1 = 3/8$; exit is free. The explicit solution, with old desired speeds 5 and 10, is

$$\rho(t,x) = \begin{cases} r_1, & 0 \le x < 5+5t, \\ 0, & 5+5t < x < 15+10t, \\ r_2, & 15+10t < x < 20+10t, \\ 0, & 20+10t < x \le 30, \end{cases} \qquad z_1(t) = 5+5t, \quad z_2(t) = 20+10t. \tag{7}$$

There are no interior lateral sources, encounters or exits in this local window. Ordinary shocks at the rear of the second block and the nonclassical zero-passing jumps at each restriction determine the same motion and field. Ahead of each restriction is vacuum, so K.A permits its held desired speed; no positive interior recovery is used. Within a block, $f(r_i)/r_i = s_i$.

The second block's left jump is an ordinary entropy shock: its speed is $(f(r_2) - 0)/(r_2 - 0) = 10$, between the triangular diagram's slopes 30 and -5. At each right jump,

$$f(r_1) - 5r_1 = 0, \qquad f(r_2) - 10r_2 = 0. \tag{8}$$

Take gauge $C(0) = m(0) = 5r_1 + 5r_2 = 5/8$. Then

$$m(t) = (5+5t)r_1 + 5r_2 = \frac{5}{8} + \frac{3}{8}t = C(t), \tag{9}$$

$$h_1(t) = C(t) - (5+5t)r_1 = \frac{1}{4}, \tag{10}$$

$$h_2(t) = C(t) - m(t) = 0. \tag{11}$$

At $t = 1/2$, positions are $15/2$ and 25, the second block begins at 20, and $m = C = 13/16$. The intervening mass between the records is exactly $1/4$. All flow, storage, position and count calculations agree in one own field.

Scope. This is an analytic local construction, not a public E1/metering trajectory or the complete sourced update. It proves more than reset interpolation, but not the full task. The exact field has two downward density jumps and cannot be represented by a single internal discontinuity. That does not exclude extending a subcell method to multiple pieces, or a qualified numerical approximation that preserves all required anchors.

3. What a genuine bridge must advance

For moving interval edges $a(t)$ and $b(t)$, conservation gives the Reynolds balance

$$\frac{d}{dt}\int_{a(t)}^{b(t)} \rho\, dx = [f - \dot{a}\rho]_{a(t)} - [f - \dot{b}\rho]_{b(t)}. \tag{12}$$

At an active zero-passing record boundary, its relative flux is zero. At a fixed transfer port, the flux comes from the same accepted node/source solve, not a second budget. This balance determines mass *after the boundary fluxes and paths have been solved*; it is not itself that solve.

For completed compatible anchors (x_i, N_i), ordered by position,

$$0 \le N_i - N_{i+1} \le \kappa(x_{i+1} - x_i), \qquad \rho_i = \frac{N_i - N_{i+1}}{x_{i+1} - x_i}. \tag{13}$$

Coincident positions must have equal count. Distinct kinetic records remain. The interpolation conserves counts and preserves vacuum when adjacent counts are equal. It does not prescribe the next moving flux or its convergence.

Source IT uses a regular-step sending mass $S_a = \Delta t D_a$, lane-choice rate π_{ab} (s^{-1}), and dimensionless oriented fraction

$$\theta_{ab} = \pi_{ab}\Delta t \quad (b \neq a), \qquad \theta_{aa} = 1 - \sum_{b\neq a} \theta_{ab}, \qquad d_{ab} = \theta_{ab} S_a. \tag{14}$$

The geometry advances (i, a) to $(i + 1, b)$ once. The source sampling, sending and available receiving define a finite numerical step. Replacing D_a by an instantaneous boundary trace while keeping the same fractions is an explicit attempted adaptation, not a proved equivalence. Internal numerical steps must not silently resample θ or E1/positive recovery.

For accepted transfer mass F_{ab}, paired balances give

$$m_a^+ = m_a + \sum_b F_{ba} - \sum_b F_{ab}. \tag{15}$$

After including outside waiting, external offers and exits, internal transfers cancel. This is conditional conservation: it does not establish the own admissibility or correctness of every F_{ab}.

K.A uses a held desired speed and the running minimum of encountered own traffic caps within a public interval, with position obtained by integration. Ending speed is retained separately. E1 attaches records to the completed own count at the next public clock. A complete numerical bridge must specify the moving/source/port solve preceding those operations. A continuous count formula supplied with external paths does not replace this requirement.

4. A direct continuous proportional-merge transplant fails consistency

This is an independent local junction diagnostic. It is *not* a reached oriented-IT/metering case and does not refute native CTM. It identifies why the step from a finite source rule to a boundary-trace rule needs qualification.

Let $d_1 = 1/5$, $d_2 = 2/5$ be demands on two independent incoming roads, $R = 3/10$ the single receiving supply, and both lane capacities $Q = 9/14$. Direct demand-proportional allocation gives

$$g_i = \frac{d_i}{d_1 + d_2} \min\{d_1 + d_2, R\}, \qquad (g_1, g_2) = \left(\frac{1}{10}, \frac{1}{5}\right). \tag{16}$$

Both roads are restricted: $g_i < d_i$. Their initial free densities are $\rho_i = d_i/u$. The admissible boundary traces for the restricted flows are congested,

$$\rho_i^* = \kappa - \frac{g_i}{w}, \qquad \rho_1^* = \frac{13}{100}, \quad \rho_2^* = \frac{11}{100}, \quad D(\rho_1^*) = D(\rho_2^*) = Q. \tag{17}$$

Each connecting shock has negative speed because

$$\sigma_i = \frac{g_i - d_i}{\rho_i^* - \rho_i} < 0. \tag{18}$$

These are appropriate upstream congestion waves. Reapplying the direct node formula to the states it generated instead returns

$$(g_1^*, g_2^*) = \left(\frac{3}{20}, \frac{3}{20}\right) \neq \left(\frac{1}{10}, \frac{1}{5}\right). \tag{19}$$

Therefore this boundary rule is not invariant under its own Riemann states. Continuous LTM literature requires a suitable invariant realization, and also distinguishes CTM interior-state interpretations. The calculation does not prove that such a source-equivalent completion is impossible.

Do not attribute this failure to N.A. With fixed priority $p = 1/2$, the selected work-conserving two-input rule is

$$g_1 = \min\{d_1, \max\{pR, R - d_2\}\}, \qquad g_2 = \min\{d_2, R - g_1\}. \tag{20}$$

For the same demands it gives $(3/20, 3/20)$, and the restricted traces with demands (Q, Q) give that same split. Another case $d_1 = 1/10$, $d_2 = 2/5$ gives $(1/10, 1/5)$, unchanged after only restricted road 2 raises its demand to Q. These checks are not a proof for all multi-input IT/moving-path nodes.

5. Qualification, not a fabricated global obstruction

Route	Established	Still required for full success
Interface	Uniform projection contradiction	A richer own field, not only renamed labels
Adaptation	Anchor consistency and the explicit two-restriction wave evolution	Complete admissible moving/node/source numerical update for the accepted phases
Representation	Continuous local count/wave construction	Endogenous paths and source-equivalent transfer ports, not externally supplied histories

An acceptable finite numerical model need not solve the continuum exactly. It must nevertheless state its own source/flux update, conditions and how it preserves the required metadata/field meaning. Ordinary field truncation error does not justify an arbitrary mismatch at retained count boundaries. Likewise a continuum method must specify how its paths and source histories are obtained from its own field. Conservation alone does not do this.

Source versus original calculation. Sections 1–2 recheck/extend the reviewer-derived local constructions under the previously selected triangular diagram and zero passing. Section 4 is an original exact diagnostic of a direct proportional-merge transplant, consistent with the source's invariance distinction. It is not presented as a literature proof against the entire approved IT graph. Section 3 records the source-time dependence and conditional balances; no new physical source rule is adopted.

Numerical alternatives. A1 retains moving anchor boundaries. A2 retains internal discontinuities inside fixed cells. C supplies a continuous count/wave reference. They can represent the same physics. Single-restriction coupled schemes and existence results are method precedents, not automatically the multiple-record, zero-passing, sampled-recovery/source task. A single-shock source formula cannot reproduce the exact two-jump field above unchanged; multi-piece extension remains possible.

Final status. No full success step; step 2 has the strongest local positive. Step 1 is refuted only on its audited domain; steps 2–3 are unestablished full constructions, not mathematical impossibility. The bounded ladder is reviewed. Q.J.F / Stage 3 / continuation 67 remain open; Stage 4 is unopened. Existing M.A, N.A, G1, E1 and K.A remain approved. Clearance, full TST response, useful gradients and necessity of retained information are not established by this companion.

Supporting sources. Full identities, access limits and extraction are in the source audit. Main junction reference: Jin, continuous LTM, section 6. The proposed fixed-cell alternative uses Chalons et al., 2017. These links are evidence provenance, not replacement update laws.